Report #5: Auditability and Security Validation in OpenStack

Bryan Daniel, Justin Cockrell, Dylan Hagy
University of South Carolina Aiken
CSCI A592-PC
Dr. Ali AlSabeh
Spring 2026

Scenario

Following the implementation of role-based access control, tenant network isolation, quota governance, and encrypted block storage, Southriver Software Solutions now operates a materially hardened OpenStack private cloud. The remaining security gap is not the absence of preventive controls, but the absence of a defined method for collecting and reviewing audit evidence that demonstrates those controls are functioning as intended [1], [3], [4].

Without an established audit workflow, administrators have limited ability to verify whether authorization decisions, network restrictions, and policy violations are being recorded in a form that supports investigation and governance. This report implements and validates a native audit review process using OpenStack administrative inspection and service logs to establish evidence collection, denied-event visibility, and anomaly identification across the existing DevStack deployment [1], [2], [4].

Prerequisites

- Functional DevStack OpenStack environment
- Implemented RBAC, network isolation, security groups, quotas, and encrypted storage
- Existing projects (admin, dev, qa) with users and roles
- Active instance (e.g., DEV-APP)
- Admin and project OpenRC files
- Root or sudo access for log inspection

Learning Objectives

- Identify OpenStack audit and logging sources
- Collect administrative evidence using OpenStack CLI
- Correlate service logs with system state
- Generate and review denied or failed actions
- Validate enforcement of RBAC, segmentation, and governance controls
- Apply a repeatable audit review workflow

Background

In smaller OpenStack environments such as DevStack, auditability is distributed across service logs and administrative state rather than provided by a single dedicated audit platform. Systemd-managed services record operational and security-relevant events through journald, while the OpenStack CLI exposes the current identity, network, and compute configuration that administrators must compare against expected policy baselines [1], [2].

This distributed evidence model is security-significant because cloud control enforcement occurs across multiple services. Keystone records identity and authorization activity, Nova records compute lifecycle behavior, and Neutron records network configuration and segmentation events. Reviewing these sources together enables administrators to validate that controls implemented in prior reports remain enforced and that failed or denied actions are observable rather than silent [2], [3].

Effective governance therefore depends not only on applying controls, but also on demonstrating that those controls produce verifiable evidence for review. A structured audit workflow improves accountability, supports anomaly detection, and mitigates the previously identified risk that policy violations or misconfigurations could persist without administrative visibility [1], [4].

Section 5.1: Baseline Audit Surface and Log Source Identification

Purpose

This section establishes the available audit and logging surface within the DevStack-based OpenStack deployment. In this environment, OpenStack services are managed through systemd and logged via journald, providing a centralized mechanism for reviewing service activity and security-relevant events. This approach aligns with OpenStack operational practices where service logs are used for troubleshooting, auditing, and validation of control enforcement [1].

Step 1: List all active DevStack service units

```
ubuntu@ubuntu:~$ systemctl list-units --type=service | grep devstack
  devstack@c-api.service                          loaded active     running       Devstack devstack@c-api.service
  devstack@c-sch.service                          loaded active     running       Devstack devstack@c-sch.service
  devstack@c-vol.service                          loaded active     running       Devstack devstack@c-vol.service
  devstack@dstat.service                          loaded active     running       Devstack devstack@dstat.service
  devstack@etcd.service                           loaded active     running       Devstack devstack@etcd.service
  devstack@g-api.service                          loaded active     running       Devstack devstack@g-api.service
  devstack@keystone.service                       loaded active     running       Devstack devstack@keystone.service
  devstack@n-api-meta.service                     loaded active     running       Devstack devstack@n-api-meta.service
  devstack@n-api.service                          loaded active     running       Devstack devstack@n-api.service
  devstack@n-cond-cell1.service                   loaded active     running       Devstack devstack@n-cond-cell1.servic
● devstack@n-cpu.service                          loaded failed     failed        Devstack devstack@n-cpu.service
  devstack@n-novnc-cell1.service                  loaded active     running       Devstack devstack@n-novnc-cell1.servi
e
  devstack@n-sch.service                          loaded active     running       Devstack devstack@n-sch.service
  devstack@n-super-cond.service                   loaded active     running       Devstack devstack@n-super-cond.servic
  devstack@neutron-api.service                    loaded active     running       Devstack devstack@neutron-api.service
  devstack@neutron-ovn-maintenance-worker.service loaded active     running       Devstack devstack@neutron-ovn-mainten
nce-worker.service
  devstack@neutron-periodic-workers.service       loaded active     running       Devstack devstack@neutron-periodic-wo
kers.service
  devstack@neutron-rpc-server.service             loaded active     running       Devstack devstack@neutron-rpc-server.
ervice
  devstack@placement-api.service                  loaded active     running       Devstack devstack@placement-api.servi
e
  devstack@q-ovn-agent.service                    loaded active     running       Devstack devstack@q-ovn-agent.service
ubuntu@ubuntu:~$
```

All the devstack services running are listed above

Step 2: Verify Nova API service status

```
ubuntu@ubuntu:~/Desktop$ systemctl status devstack@n-api.service
● devstack@n-api.service - Devstack devstack@n-api.service
     Loaded: loaded (/etc/systemd/system/devstack@n-api.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-27 10:43:09 EDT; 7min ago
   Main PID: 954 (uwsgi)
     Status: "uWSGI is ready"
      Tasks: 3 (limit: 9199)
     Memory: 254.9M (peak: 256.4M)
        CPU: 15.419s
     CGroup: /system.slice/system-devstack.slice/devstack@n-api.service
             ├─954 "nova-apiuWSGI master"
             ├─1200 "nova-apiuWSGI worker 1"
             └─1201 "nova-apiuWSGI worker 2"

Mar 27 10:43:56 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082>
Mar 27 10:43:56 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WSGI app 0 (mountpoint='') ready in 48 seconds on interpreter 0x7c>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c>
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WSGI app 0 (mountpoint='') ready in 48 seconds on interpreter 0x7c>
lines 1-23/23 (END)
```

The Nova API service is active

Step 3: Verify Neutron server service status

```
ubuntu@ubuntu:~/Desktop$ systemctl status devstack@neutron-api.service
● devstack@neutron-api.service - Devstack devstack@neutron-api.service
     Loaded: loaded (/etc/systemd/system/devstack@neutron-api.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-27 10:43:09 EDT; 10min ago
   Main PID: 960 (uwsgi)
     Status: "uWSGI is ready"
      Tasks: 59 (limit: 9199)
     Memory: 313.9M (peak: 337.3M)
        CPU: 26.583s
     CGroup: /system.slice/system-devstack.slice/devstack@neutron-api.service
             ├─960 "neutron-apiuWSGI master"
             ├─1192 "neutron-apiuWSGI worker 1"
             └─1199 "neutron-apiuWSGI worker 2"

Mar 27 10:52:59 ubuntu devstack@neutron-api.service[1199]: DEBUG futurist.periodics [-] Submitting periodic callback 'n>
Mar 27 10:52:59 ubuntu devstack@neutron-api.service[1199]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.main>
Mar 27 10:52:59 ubuntu devstack@neutron-api.service[1192]: DEBUG futurist.periodics [-] Submitting periodic callback 'n>
Mar 27 10:52:59 ubuntu devstack@neutron-api.service[1192]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.main>
Mar 27 10:53:09 ubuntu devstack@neutron-api.service[1199]: DEBUG dbcounter [-] [1199] Writing DB stats neutron:UPDATE=1>
Mar 27 10:53:09 ubuntu devstack@neutron-api.service[1192]: DEBUG dbcounter [-] [1192] Writing DB stats neutron:UPDATE=1>
Mar 27 10:53:14 ubuntu devstack@neutron-api.service[1199]: DEBUG futurist.periodics [-] Submitting periodic callback 'n>
Mar 27 10:53:14 ubuntu devstack@neutron-api.service[1199]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.main>
Mar 27 10:53:14 ubuntu devstack@neutron-api.service[1192]: DEBUG futurist.periodics [-] Submitting periodic callback 'n>
Mar 27 10:53:14 ubuntu devstack@neutron-api.service[1192]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.main>
lines 1-23/23 (END)
```

The Neutron server is active

Step 4: Verify Neutron agent service status

```
ubuntu@ubuntu:~/Desktop$ systemctl status devstack@q-ovn-agent.service
● devstack@q-ovn-agent.service - Devstack devstack@q-ovn-agent.service
     Loaded: loaded (/etc/systemd/system/devstack@q-ovn-agent.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-27 10:43:06 EDT; 12min ago
   Main PID: 965 (neutron-ovn-age)
      Tasks: 16 (limit: 9199)
     Memory: 311.9M (peak: 313.2M)
        CPU: 9.052s
     CGroup: /system.slice/system-devstack.slice/devstack@q-ovn-agent.service
             ├─965 "neutron-ovn-agent: master process [/opt/stack/data/venv/bin/neutron-ovn-agent --config-file /etc/n>
             ├─3461 "neutron-ovn-agent: ServiceWrapper worker(0)"
             └─4287 /opt/stack/data/venv/bin/python3.12 /usr/local/bin/privsep-helper --config-file /etc/neutron/plugin>

Mar 27 10:53:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" "rele>
Mar 27 10:53:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Acquiring lock "_check_child_proces>
Mar 27 10:53:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" acqui>
Mar 27 10:53:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" "rele>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Acquiring lock "_check_child_proces>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" acqui>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" "rele>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Acquiring lock "_check_child_proces>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" acqui>
Mar 27 10:54:31 ubuntu neutron-ovn-agent[965]: DEBUG oslo_concurrency.lockutils [-] Lock "_check_child_processes" "rele>
lines 1-22/22 (END)
```

The Neutron agent service is active

Step 5: Verify Keystone service status

```
ubuntu@ubuntu:~/Desktop$ systemctl status devstack@keystone.service
● devstack@keystone.service - Devstack devstack@keystone.service
     Loaded: loaded (/etc/systemd/system/devstack@keystone.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-27 10:43:09 EDT; 14min ago
   Main PID: 952 (uwsgi)
     Status: "uWSGI is ready"
      Tasks: 3 (limit: 9199)
     Memory: 210.1M (peak: 216.6M)
        CPU: 12.940s
     CGroup: /system.slice/system-devstack.slice/devstack@keystone.service
             ├─952 "keystoneuWSGI master"
             ├─1189 "keystoneuWSGI worker 1"
             └─1190 "keystoneuWSGI worker 2"

Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes t>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: WSGI app 0 (mountpoint='') ready in 47 seconds on interpreter 0>
Mar 27 10:43:56 ubuntu devstack@keystone.service[1189]: WSGI app 0 (mountpoint='') ready in 47 seconds on interpreter 0>
lines 1-23/23 (END)
```

The Keystone service is active

Step 6: Verify Glance API service status

```
ubuntu@ubuntu:~/Desktop$ systemctl status devstack@g-api.service
● devstack@g-api.service - Devstack devstack@g-api.service
     Loaded: loaded (/etc/systemd/system/devstack@g-api.service; enabled; preset: enabled)
     Active: active (running) since Fri 2026-03-27 10:43:09 EDT; 15min ago
   Main PID: 951 (uwsgi)
     Status: "uWSGI is ready"
      Tasks: 7 (limit: 9199)
     Memory: 229.4M (peak: 233.0M)
        CPU: 11.414s
     CGroup: /system.slice/system-devstack.slice/devstack@g-api.service
             ├─951 "glance-apiuWSGI master"
             ├─1195 "glance-apiuWSGI worker 1"
             └─1196 "glance-apiuWSGI worker 2"

Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: DEBUG glance.common.config [None req-7ce17f87-3b97-4210-b478-cc49f>
Mar 27 10:43:55 ubuntu devstack@g-api.service[1195]: WSGI app 0 (mountpoint='') ready in 46 seconds on interpreter 0x7f>
Mar 27 10:44:03 ubuntu devstack@g-api.service[1196]: DEBUG dbcounter [-] [1196] Writing DB stats glance:INSERT=4 {{(pid>
Mar 27 10:44:04 ubuntu devstack@g-api.service[1195]: DEBUG dbcounter [-] [1195] Writing DB stats glance:INSERT=4 {{(pid>
lines 1-23/23 (END)
```

The Glance API service is active

Step 7: Review recent Nova API log entries with the command: journalctl –u devstack@n-api.service --no-pager -n 50

```
Mar 27 10:43:56 ubuntu devstack@n-api.service[1201]: DEBUG oslo_concurrency.processutils [None req-0978de6c-45e6-412c-9d
f5-15a8b233fe69 None None] CMD "grep -F node.session.scan /sbin/iscsiadm" returned: 0 in 0.019s {{{(pid=1201) execute /op
t/stack/data/venv/lib/python3.12/site-packages/oslo_concurrency/processutils.py:372}}
Mar 27 10:43:56 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082-
ba97-76e1e9a0fac4 None None] AuthToken middleware is set with keystone_authtoken.service_token_roles_required set to Fal
se. This is backwards compatible but deprecated behaviour. Please set this to True.
Mar 27 10:43:56 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082-
ba97-76e1e9a0fac4 None None] Configuring www_authenticate_uri to point to the public identity endpoint is required; clie
nts may not be able to authenticate against an admin endpoint
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c-
9df5-15a8b233fe69 None None] AuthToken middleware is set with keystone_authtoken.service_token_roles_required set to Fal
se. This is backwards compatible but deprecated behaviour. Please set this to True.
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c-
9df5-15a8b233fe69 None None] Configuring www_authenticate_uri to point to the public identity endpoint is required; clie
nts may not be able to authenticate against an admin endpoint
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082-
ba97-76e1e9a0fac4 None None] AuthToken middleware is set with keystone_authtoken.service_token_roles_required set to Fal
se. This is backwards compatible but deprecated behaviour. Please set this to True.
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WARNING keystonemiddleware.auth_token [None req-c557f67b-77c7-4082-
ba97-76e1e9a0fac4 None None] Configuring www_authenticate_uri to point to the public identity endpoint is required; clie
nts may not be able to authenticate against an admin endpoint
Mar 27 10:43:57 ubuntu devstack@n-api.service[1200]: WSGI app 0 (mountpoint='') ready in 48 seconds on interpreter 0x7dd
0ebca3668 pid: 1200 (default app)
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c-
9df5-15a8b233fe69 None None] AuthToken middleware is set with keystone_authtoken.service_token_roles_required set to Fal
se. This is backwards compatible but deprecated behaviour. Please set this to True.
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WARNING keystonemiddleware.auth_token [None req-0978de6c-45e6-412c-
9df5-15a8b233fe69 None None] Configuring www_authenticate_uri to point to the public identity endpoint is required; clie
nts may not be able to authenticate against an admin endpoint
Mar 27 10:43:57 ubuntu devstack@n-api.service[1201]: WSGI app 0 (mountpoint='') ready in 48 seconds on interpreter 0x7dd
0ebca3668 pid: 1201 (default app)
ubuntu@ubuntu:~/Desktop$
```

Now we can see the Nova API logs. These show:

- Keystone authentication warnings about deprecated auth_token middleware configuration
- WSGI app initialization messages showing the Nova API starting up
- Configuration warnings about authentication settings

These logs provide audit evidence of API service startup and configuration validation.

Step 8: Review recent Neutron server log entries with the command: journalctl -u devstack@neutron-api.service --no-pager -n 50

```
utron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maintenance.HashRingHealthCheckPeriodics.touch_hash_ring_node' {{(pid=11
92) _process_scheduled /opt/stack/data/venv/lib/python3.12/site-packages/futurist/periodics.py:638}}
Mar 27 11:04:29 ubuntu devstack@neutron-api.service[1192]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maint
enance [-] Touching Hash Ring node "1540472e2aba5f63af8c71b6f352899d" from periodic health check thread {{(pid=1192) tou
ch_hash_ring_node /opt/stack/neutron/neutron/plugins/ml2/drivers/ovn/mech_driver/ovsdb/maintenance.py:1216}}
Mar 27 11:04:39 ubuntu devstack@neutron-api.service[1199]: DEBUG dbcounter [-] [1199] Writing DB stats neutron:UPDATE=1,
neutron:SELECT=1 {{(pid=1199) stat_writer /opt/stack/data/venv/lib/python3.12/site-packages/dbcounter.py:115}}
Mar 27 11:04:39 ubuntu devstack@neutron-api.service[1192]: DEBUG dbcounter [-] [1192] Writing DB stats neutron:UPDATE=1,
neutron:SELECT=1 {{(pid=1192) stat_writer /opt/stack/data/venv/lib/python3.12/site-packages/dbcounter.py:115}}
Mar 27 11:04:44 ubuntu devstack@neutron-api.service[1199]: DEBUG futurist.periodics [-] Submitting periodic callback 'ne
utron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maintenance.HashRingHealthCheckPeriodics.touch_hash_ring_node' {{(pid=11
99) _process_scheduled /opt/stack/data/venv/lib/python3.12/site-packages/futurist/periodics.py:638}}
Mar 27 11:04:44 ubuntu devstack@neutron-api.service[1199]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maint
enance [-] Touching Hash Ring node "9c259ccce132591992df8b4de0200817" from periodic health check thread {{(pid=1199) tou
ch_hash_ring_node /opt/stack/neutron/neutron/plugins/ml2/drivers/ovn/mech_driver/ovsdb/maintenance.py:1216}}
Mar 27 11:04:44 ubuntu devstack@neutron-api.service[1192]: DEBUG futurist.periodics [-] Submitting periodic callback 'ne
utron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maintenance.HashRingHealthCheckPeriodics.touch_hash_ring_node' {{(pid=11
92) _process_scheduled /opt/stack/data/venv/lib/python3.12/site-packages/futurist/periodics.py:638}}
Mar 27 11:04:44 ubuntu devstack@neutron-api.service[1192]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maint
enance [-] Touching Hash Ring node "1540472e2aba5f63af8c71b6f352899d" from periodic health check thread {{(pid=1192) tou
ch_hash_ring_node /opt/stack/neutron/neutron/plugins/ml2/drivers/ovn/mech_driver/ovsdb/maintenance.py:1216}}
Mar 27 11:04:54 ubuntu devstack@neutron-api.service[1199]: DEBUG dbcounter [-] [1199] Writing DB stats neutron:UPDATE=1,
neutron:SELECT=1 {{(pid=1199) stat_writer /opt/stack/data/venv/lib/python3.12/site-packages/dbcounter.py:115}}
Mar 27 11:04:54 ubuntu devstack@neutron-api.service[1192]: DEBUG dbcounter [-] [1192] Writing DB stats neutron:UPDATE=1,
neutron:SELECT=1 {{(pid=1192) stat_writer /opt/stack/data/venv/lib/python3.12/site-packages/dbcounter.py:115}}
Mar 27 11:04:59 ubuntu devstack@neutron-api.service[1199]: DEBUG futurist.periodics [-] Submitting periodic callback 'ne
utron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maintenance.HashRingHealthCheckPeriodics.touch_hash_ring_node' {{(pid=11
99) _process_scheduled /opt/stack/data/venv/lib/python3.12/site-packages/futurist/periodics.py:638}}
Mar 27 11:04:59 ubuntu devstack@neutron-api.service[1199]: DEBUG neutron.plugins.ml2.drivers.ovn.mech_driver.ovsdb.maint
enance [-] Touching Hash Ring node "9c259ccce132591992df8b4de0200817" from periodic health check thread {{(pid=1199) tou
ch_hash_ring_node /opt/stack/neutron/neutron/plugins/ml2/drivers/ovn/mech_driver/ovsdb/maintenance.py:1216}}
ubuntu@ubuntu:~/Desktop$
```

The Neutron logs show:

- OVN mechanism driver activity with periodic health checks
- Hash ring node operations for distributed networking
- Database updates (neutron:UPDATE operations)
- Periodic callbacks for network maintenance

These logs demonstrate active network management and provide audit trails for network operations.

Step 9: Review recent Keystone-related log entries with the command: journalctl -u devstack@keystone.service --no-pager -n 50

```
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/credentials/OS-EC2/<string:credential_id>' {}] {{(pid=1190) _add_mapped_resources
/opt/stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/OS-OAUTH1/access_tokens' {}] {{(pid=1190) _add_mapped_resources /opt/stack/keyston
e/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/OS-OAUTH1/access_tokens/<string:access_token_id>' {}] {{(pid=1190) _add_mapped_res
ources /opt/stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/OS-OAUTH1/access_tokens/<string:access_token_id>/roles' {}] {{(pid=1190) _add_mapp
ed_resources /opt/stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/OS-OAUTH1/access_tokens/<string:access_token_id>/roles/<string:role_id>' {}] {{(pi
d=1190) _add_mapped_resources /opt/stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/application_credentials' {}] {{(pid=1190) _add_mapped_resources /opt/stack/keyston
e/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/application_credentials/<string:application_credential_id>' {}] {{(pid=1190) _add_
mapped_resources /opt/stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/access_rules' {}] {{(pid=1190) _add_mapped_resources /opt/stack/keystone/keystone/
server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: DEBUG keystone.server.flask.common [-] Adding resource routes to
 API users: ['/users/<string:user_id>/access_rules/<string:access_rule_id>' {}] {{(pid=1190) _add_mapped_resources /opt/
stack/keystone/keystone/server/flask/common.py:483}}
Mar 27 10:43:56 ubuntu devstack@keystone.service[1190]: WSGI app 0 (mountpoint='') ready in 47 seconds on interpreter 0x
7b40abca3668 pid: 1190 (default app)
Mar 27 10:43:56 ubuntu devstack@keystone.service[1189]: WSGI app 0 (mountpoint='') ready in 47 seconds on interpreter 0x
7b40abca3668 pid: 1189 (default app)
ubuntu@ubuntu:~/Desktop$
```

Keystone logs show:

- Resource route additions for API endpoints (users, credentials, tokens, roles, access rules)
- WSGI application initialization on multiple interpreters
- Flask framework routing for identity management

These logs provide the foundation for authentication audit trails.

Section 5.1 Complete

You've successfully mapped the audit surface. You have evidence that all critical OpenStack services are running and generating logs through systemd/journald.

Key outcomes for Section 5.1:

- Identified 20+ active DevStack services
- Verified health of Nova API, Neutron API, OVN agent, Keystone, and Glance
- Demonstrated log collection from each service via journalctl
- Established baseline for audit monitoring

Section 5.2: Collection of Security-Relevant Audit Evidence from Native Services

Purpose

OpenStack services generate logs that capture authentication, authorization, resource provisioning, and network operations. These logs provide administrators with audit evidence necessary to validate enforcement of cloud security controls such as role-based access control (RBAC), network segmentation, and tenant isolation [2].

This section establishes an administrative evidence baseline by enumerating identities, projects, networks, routers, security groups, and instance state, then correlating that state with native Nova and Neutron logging. The objective is to validate that previously implemented security controls are not only configured, but also observable through records that support review, accountability, and incident reconstruction.

Step 1: List all users and roles

```
ubuntu@ubuntu:~/Desktop$ openstack user list
+----------------------------------+-----------------+
| ID                               | Name            |
+----------------------------------+-----------------+
| 0526a36658ec449e9dc865112109a48b | placement       |
| 2208429300c84a71a2318390439f7495 | Auditor1        |
| 22f93e8d2cbe4128814c36d3be38bbee | system_reader   |
| 2ecde72a934244baa49e47aae373498d | system_member   |
| 2eee114ca75f4851850324b95989f70f | admin           |
| 45cfc821007147aa9beca67e4c54d37c | cinder          |
| 58eb5834ea50412880f0a51f259834c6 | neutron         |
| 7e573f39bdc54bf9ab6c63524c265a36 | alt_demo_member |
| 86085e7f4bf849d58c45534c777fa5e9 | nova            |
| 95505b2e82304122828c763604842a38 | glance          |
| a8050f3abec64a64b8669b536550b798 | alt_demo        |
| ba8470eaec4e4fe58738e1649af394cd | demo            |
| c9bf6df3c8d64124a00735dc6bd5876f | demo_reader     |
| dcf040f6fbc04e3f82e97f8b9a8fa38a | alt_demo_reader |
| f25bd4eea4034da9b65d156c48a50cd2 | DevUser1        |
| fa4ebda00afc4997bb8bb81570313267 | QAUser1         |
+----------------------------------+-----------------+
```

Users are displayed

```
ubuntu@ubuntu:~/Desktop$ openstack role list
+----------------------------------+---------------+
| ID                               | Name          |
+----------------------------------+---------------+
| 0d37478aaf6547ca9431a61875ac4581 | service       |
| 1dea663380214c78ac6299578e9aa902 | audit_read    |
| 20dd4f94259549b39a030d4a47b21887 | dev_compute   |
| 2bbb2fc206bb4097bd9a9584433b3251 | reader        |
| 2c88f95e06554c44ad4c61e238b6d830 | member        |
| 4577276bcc824386b2189c61e7c0d70e | qa_compute    |
| 525553ab65ab4822a2cfa7b16e5e849a | manager       |
| 7f62fa9295294cfba8540fedf74a8478 | ResellerAdmin |
| 8dba605dda2f4199bac4e12478c0979e | anotherrole   |
| b6cebb4885fe4b678d2159db040ce296 | admin         |
+----------------------------------+---------------+
```

All Roles are displayed

Step 2: List all projects

```
ubuntu@ubuntu:~/Desktop$ openstack project list
+----------------------------------+--------------------+
| ID                               | Name               |
+----------------------------------+--------------------+
| 4f97f0eebec74d8b83b9296bc70dd0bb | dev                |
| 513076bc0448439a88fda9436d9078e5 | alt_demo           |
| 7439add45b594797a6de39131ab05cc8 | invisible_to_admin |
| 7e6bf84624f54b56b4da0f27b6c16aed | qa                 |
| bd755daf91444d328d3f25316317a883 | admin              |
| ecfddf1248db4233aad57146f0b0b792 | demo               |
| fd2553c6119d4c41aae1a732d85d9107 | service            |
+----------------------------------+--------------------+
```

All Projects are displayed

Step 3: List networks to establish baseline segmentation

```
ubuntu@ubuntu:~/Desktop$ openstack network list
+--------------------------------------+----------+----------------------------------------------------------------+
| ID                                   | Name     | Subnets                                                        |
+--------------------------------------+----------+----------------------------------------------------------------+
| 24353f0c-c770-4e7a-abd7-196dcb3bc476 | public   | 0fd39408-8388-411b-9d1a-f564cfd67967,                          |
|                                      |          | 95515952-8b36-4743-924d-21b00e17e63c                           |
| 398b5405-9388-4b39-9937-8e855b33c735 | shared   | fd117f77-351a-4a10-a454-beb9919beca3                           |
| a729b745-9ead-4256-b0b5-1ce16ef6d2f7 | dev-net  | 9a8c2511-facf-4d12-bdaf-cb5a33ef1bec                           |
| d3e3e853-ec24-43aa-969b-1d7dfb5e9f63 | test-net | 76cc7d70-112c-488d-acb8-5d1f99c5ca14                           |
| e7eefaea-083b-4004-bf10-1aa9c76172d5 | private  | f77829e9-5891-42f9-a3bf-689666dc902c,                          |
|                                      |          | fa40244b-2d8b-4e48-bbce-a23a60a240c1                           |
+--------------------------------------+----------+----------------------------------------------------------------+
```

All networks are displayed

Step 4: List subnets associated with networks

```
ubuntu@ubuntu:~/Desktop$ openstack subnet list
+--------------------------------------+----------------------+--------------------------------------+--------------------+
| ID                                   | Name                 | Network                              | Subnet             |
+--------------------------------------+----------------------+--------------------------------------+--------------------+
| 0fd39408-8388-411b-9d1a-             | ipv6-public-subnet   | 24353f0c-c770-4e7a-abd7-             | 2001:db8::/64      |
| f564cfd67967                         |                      | 196dcb3bc476                         |                    |
| 76cc7d70-112c-488d-acb8-             | test-subnet          | d3e3e853-ec24-43aa-969b-             | 10.0.10.0/24       |
| 5d1f99c5ca14                         |                      | 1d7dfb5e9f63                         |                    |
| 95515952-8b36-4743-924d-             | public-subnet        | 24353f0c-c770-4e7a-abd7-             | 172.24.4.0/24      |
| 21b00e17e63c                         |                      | 196dcb3bc476                         |                    |
| 9a8c2511-facf-4d12-bdaf-             | dev-subnet           | a729b745-9ead-4256-b0b5-             | 10.0.20.0/24       |
| cb5a33ef1bec                         |                      | 1ce16ef6d2f7                         |                    |
| f77829e9-5891-42f9-a3bf-             | private-subnet       | e7eefaea-083b-4004-bf10-             | 10.0.0.0/26        |
| 689666dc902c                         |                      | 1aa9c76172d5                         |                    |
| fa40244b-2d8b-4e48-bbce-             | ipv6-private-subnet  | e7eefaea-083b-4004-bf10-             | fd59:3f9e:7361::/64 |
| a23a60a240c1                         |                      | 1aa9c76172d5                         |                    |
| fd117f77-351a-4a10-a454-             | shared-subnet        | 398b5405-9388-4b39-9937-             | 192.168.233.0/24   |
| beb9919beca3                         |                      | 8e855b33c735                         |                    |
+--------------------------------------+----------------------+--------------------------------------+--------------------+
```

All subnets are displayed

Step 5: List routers and external connectivity

```
ubuntu@ubuntu:~/Desktop$ openstack router list
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
| ID                                   | Name        | Status | State | Project                          | HA   |
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
| 077bdf3b-2cd5-45d3-9156-8a5aaa4704d5 | dev-router  | ACTIVE | UP    | 4f97f0eebec74d8b83b9296bc70dd0bb | True |
| 1ab77af1-4929-4525-8c72-1da8453f7bbc | router1     | ACTIVE | UP    | ecfddf1248db4233aad57146f0b0b792 | True |
| d1019b36-f21b-490b-84ae-3c6ccb51b52d | test-router | ACTIVE | UP    | 7e6bf84624f54b56b4da0f27b6c16aed | True |
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
```

All routers are displayed

Step 6: Show detailed information for a router

```
ubuntu@ubuntu:~/Desktop$ openstack router show router1
+-------------------------+-----------------------------------------------------------------------------------------+
| Field                   | Value                                                                                   |
+-------------------------+-----------------------------------------------------------------------------------------+
| admin_state_up          | UP                                                                                      |
| availability_zone_hints |                                                                                         |
| availability_zones      |                                                                                         |
| created_at              | 2025-09-12T23:47:01Z                                                                    |
| description             |                                                                                         |
| enable_default_route_bfd | False                                                                                   |
| enable_default_route_ecmp | False                                                                                   |
| enable_ndp_proxy        | None                                                                                    |
| external_gateway_info   | {"network_id": "24353f0c-c770-4e7a-abd7-196dcb3bc476", "external_fixed_ips":            |
|                         | [{"subnet_id": "95515952-8b36-4743-924d-21b00e17e63c", "ip_address": "172.24.4.6"},     |
|                         | {"subnet_id": "0fd39408-8388-411b-9d1a-f564cfd67967", "ip_address": "2001:db8::1"}],    |
|                         | "enable_snat": true}                                                                    |
| external_gateways       | [{'network_id': '24353f0c-c770-4e7a-abd7-196dcb3bc476', 'external_fixed_ips':           |
|                         | [{'ip_address': '172.24.4.6', 'subnet_id': '95515952-8b36-4743-924d-21b00e17e63c'},     |
|                         | {'ip_address': '2001:db8::1', 'subnet_id': '0fd39408-8388-411b-9d1a-f564cfd67967'}]}]   |
| flavor_id               | None                                                                                    |
| ha                      | True                                                                                    |
| id                      | 1ab77af1-4929-4525-8c72-1da8453f7bbc                                                    |
| interfaces_info         | [{"port_id": "440df7a3-9c04-470f-ac58-6d3413cf22fc", "ip_address": "fd59:3f9e:7361::1",  |
|                         | "subnet_id": "fa40244b-2d8b-4e48-bbce-a23a60a240c1"}, {"port_id":                       |
|                         | "c4e8d001-5901-4f5c-85a4-0348d8835640", "ip_address": "10.0.0.1", "subnet_id":          |
|                         | "f77829e9-5891-42f9-a3bf-689666dc902c"}]                                                |
| name                    | router1                                                                                 |
| project_id              | ecfddf1248db4233aad57146f0b0b792                                                        |
| revision_number         | 4                                                                                       |
| routes                  |                                                                                         |
| status                  | ACTIVE                                                                                  |
| tags                    |                                                                                         |
| updated_at              | 2025-09-12T23:47:14Z                                                                    |
+-------------------------+-----------------------------------------------------------------------------------------+
```

The config for Router1 Is displayed

Step 7: List security groups

```
ubuntu@ubuntu:~/Desktop$ openstack security group list
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
| ID                                   | Name        | Description                 | Project                          | Tags | Shared |
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
| 1c53e01b-9b6b-431b-8b12-964836832e18 | test-secure | TEST inbound: SSH, ICMP only | 7e6bf84624f54b56b4da0f27b6c16aed | []   | False  |
| 278ac93e-b084-418e-8636-426c452fefd1 | dev-secure  | DEV inbound: SSH, ICMP, HTTP | 4f97f0eebec74d8b83b9296bc70dd0bb | []   | False  |
| 3ca27139-4e0d-41fa-9c43-9f8ee7f0312d | default     | Default security group      | ecfddf1248db4233aad57146f0b0b792 | []   | False  |
| 8fb9e636-2eae-4b42-a99c-84ea50dbb2c0 | default     | Default security group      | fd2553c6119d4c41aae1a732d85d9107 | []   | False  |
| a6f23c36-121e-4c3e-9a7f-fc1870b81aef | default     | Default security group      | 4f97f0eebec74d8b83b9296bc70dd0bb | []   | False  |
| ac040561-2ce1-43fb-8f5f-6551f53fd301 | default     | Default security group      | 7e6bf84624f54b56b4da0f27b6c16aed | []   | False  |
| ba8bc8c6-7f60-4d7f-9a0c-1b8870315774 | default     | Default security group      | bd755daf91444d328d3f25316317a883 | []   | False  |
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
```

All Security Groups are displayed

Step 8: Show rules for a specific security group

```
ubuntu@ubuntu:~/Desktop$ openstack security group rule list test-secure
+--------------------------------------+-------------+-----------+-----------+------------+-----------+-----------------------+----------------------+
| ID                                   | IP Protocol | Ethertype | IP Range  | Port Range | Direction | Remote Security Group | Remote Address Group |
+--------------------------------------+-------------+-----------+-----------+------------+-----------+-----------------------+----------------------+
| 9c2ac664-be87-4153-918e-5d394d67392b | icmp        | IPv4      | 0.0.0.0/0 |            | ingress   | None                  | None                 |
| b149eb85-e722-4670-a4dd-efbbc60a37cb | tcp         | IPv4      | 0.0.0.0/0 | 22:22      | ingress   | None                  | None                 |
| b6474a16-31bb-4646-980b-9bfd5a0eb8df | None        | IPv4      | 0.0.0.0/0 |            | egress    | None                  | None                 |
| c9be8053-d790-476d-bb72-eef181cc921b | None        | IPv6      | ::/0      |            | egress    | None                  | None                 |
+--------------------------------------+-------------+-----------+-----------+------------+-----------+-----------------------+----------------------+
```

A few catch all ACLs are configured

Step 9: List instances across all projects (admin context)

```
ubuntu@ubuntu:~/Desktop$ openstack server list --all-projects
+--------------------------+-------------------------+---------+-------------------------+-------------------------+---------+-----------------------------+
| ID                       | Name                    | Status  | Networks                | Image                   | Flavor  | Project ID                  |
+--------------------------+-------------------------+---------+-------------------------+-------------------------+---------+-----------------------------+
| 6dd37da6-1bc3-4364-      | DEV-APP                 | ACTIVE  | dev-net=10.0.20.86      | cirros-0.6.3-x86_64-    | m1.tiny | 4f97f0eebec74d8b83b9296bc7  |
| 856f-510968779773        |                         |         |                         | disk                    |         | 0dd0bb                      |
| d0580bec-717d-40ac-      | governance-test-        | SHUTOFF | dev-net=10.0.20.71      | cirros-0.6.3-x86_64-    | m1.tiny | 4f97f0eebec74d8b83b9296bc7  |
| 877e-99509090437f        | instance                |         |                         | disk                    |         | 0dd0bb                      |
| ec9d8b0f-b1b0-4166-      | TEST-DB                 | SHUTOFF | test-net=10.0.10.195,   | N/A (booted from        | m1.tiny | 7e6bf84624f54b56b4da0f27b6  |
| a0e7-60688f7bb11c        |                         |         | 172.24.4.107            | volume)                 |         | c16aed                      |
| 0925a67e-0b77-4536-      | TEST-APP                | SHUTOFF | test-net=10.0.10.184,   | N/A (booted from        | m1.tiny | 7e6bf84624f54b56b4da0f27b6  |
| 8789-5f8207605940        |                         |         | 172.24.4.18             | volume)                 |         | c16aed                      |
| f45bc2de-808d-46ab-      | DEV-DB                  | SHUTOFF | dev-net=10.0.20.126,    | N/A (booted from        | m1.tiny | 4f97f0eebec74d8b83b9296bc7  |
| bce3-294d37c939bb        |                         |         | 172.24.4.78             | volume)                 |         | 0dd0bb                      |
+--------------------------+-------------------------+---------+-------------------------+-------------------------+---------+-----------------------------+
```

All servers are displayed

Step 10: Review Nova logs for instance lifecycle activity with the command: journalctl -u devstack@n-api.service | grep -i "instance"

```
Feb 12 14:10:28 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:10:28 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:10:28 ubuntu devstack@n-api.service[1195]: DEBUG nova.compute.api [None req-17b7c0ba-4888-47cf-8354-dc576232dd65 dev admin] [instance: e307b
644-e163-40c6-b90e-c583e5ff1055] Fetching instance by UUID {{(pid=1195) get /opt/stack/nova/nova/compute/api.py:3017}}
Feb 12 14:11:18 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:11:18 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:11:18 ubuntu devstack@n-api.service[1196]: DEBUG nova.compute.api [None req-f6e85ffa-8ab2-432f-991d-98e611b45ded dev admin] [instance: e307b
644-e163-40c6-b90e-c583e5ff1055] Fetching instance by UUID {{(pid=1196) get /opt/stack/nova/nova/compute/api.py:3017}}
Feb 12 14:11:28 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:11:28 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:11:29 ubuntu devstack@n-api.service[1195]: DEBUG nova.compute.api [None req-1a2f1580-c1b0-45fa-acae-80ad6342f4c2 dev admin] [instance: e307b
644-e163-40c6-b90e-c583e5ff1055] Fetching instance by UUID {{(pid=1195) get /opt/stack/nova/nova/compute/api.py:3017}}
Feb 12 14:12:18 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:12:18 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:12:18 ubuntu devstack@n-api.service[1196]: DEBUG nova.compute.api [None req-bee687e2-fec4-4468-b54e-6d432cccdaf1 dev admin] [instance: e307b
644-e163-40c6-b90e-c583e5ff1055] Fetching instance by UUID {{(pid=1196) get /opt/stack/nova/nova/compute/api.py:3017}}
Feb 12 14:12:29 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     output = handler(context, instance, console_type)
Feb 12 14:12:29 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:12:29 ubuntu devstack@n-api.service[1195]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:13:19 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:13:19 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
Feb 12 14:13:19 ubuntu devstack@n-api.service[1196]: DEBUG nova.compute.api [None req-79e04fc5-d242-41ef-8421-c60c670e7439 dev admin] [instance: e307b
644-e163-40c6-b90e-c583e5ff1055] Fetching instance by UUID {{(pid=1196) get /opt/stack/nova/nova/compute/api.py:3017}}
Feb 12 14:14:19 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     output = handler(context, instance, console_type)
Feb 12 14:14:19 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return function(self, context, instance, *args, **kwargs)
Feb 12 14:14:19 ubuntu devstack@n-api.service[1196]: ERROR nova.api.openstack.wsgi     return f(self, context, instance, *args, **kw)
```

A few errors are found in these logs.

Step 11: Review Neutron logs for network-related activity with the command: journalctl -u devstack@n-api.service | grep -i "network"

```
550}}
Feb 11 20:18:10 ubuntu devstack@n-api.service[1133]: INFO nova.api.openstack.compute.server_external_events [req-93b27118-24f9-46b4-8db8-ef21d2909a4d
req-fa501342-1b5f-4229-9d31-8b34b83c22a1 service nova] Creating event network-changed:1b396c61-7597-44fd-ad96-23a6090ff468 for instance ec9d8b0f-b1b0-
4166-a0e7-60688f7bb11c on ubuntu
Feb 11 20:18:32 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi [req-69eba19e-41f6-49d3-9325-a98270b657ab req-a89b5d7c-1c9e-41de-8a
83-b1213ad863f9 service nova] Action: 'create', calling method: <bound method ServerExternalEventsController.create of <nova.api.openstack.compute.ser
ver_external_events.ServerExternalEventsController object at 0x75af14392120>>, body: {"events": [{"server_uuid": "ec9d8b0f-b1b0-4166-a0e7-60688f7bb11c
", "name": "network-vif-plugged", "status": "completed", "tag": "1b396c61-7597-44fd-ad96-23a6090ff468"}]} {{(pid=1132) _process_stack /opt/stack/nova/
nova/api/openstack/wsgi.py:550}}
Feb 11 20:18:32 ubuntu devstack@n-api.service[1132]: INFO nova.api.openstack.compute.server_external_events [req-69eba19e-41f6-49d3-9325-a98270b657ab
req-a89b5d7c-1c9e-41de-8a83-b1213ad863f9 service nova] Creating event network-vif-plugged:1b396c61-7597-44fd-ad96-23a6090ff468 for instance ec9d8b0f-b
1b0-4166-a0e7-60688f7bb11c on ubuntu
Feb 11 20:18:34 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi [req-40d21a23-362e-482f-adac-330548462e35 req-b9b81dfb-f684-4dc7-be
98-fa0c368a5e93 service nova] Action: 'create', calling method: <bound method ServerExternalEventsController.create of <nova.api.openstack.compute.ser
ver_external_events.ServerExternalEventsController object at 0x75af14491df0>>, body: {"events": [{"server_uuid": "ec9d8b0f-b1b0-4166-a0e7-60688f7bb11c
", "name": "network-vif-plugged", "status": "completed", "tag": "1b396c61-7597-44fd-ad96-23a6090ff468"}]} {{(pid=1133) _process_stack /opt/stack/nova/
nova/api/openstack/wsgi.py:550}}
Feb 11 20:18:34 ubuntu devstack@n-api.service[1133]: INFO nova.api.openstack.compute.server_external_events [req-40d21a23-362e-482f-adac-330548462e35
req-b9b81dfb-f684-4dc7-be98-fa0c368a5e93 service nova] Creating event network-vif-plugged:1b396c61-7597-44fd-ad96-23a6090ff468 for instance ec9d8b0f-b
1b0-4166-a0e7-60688f7bb11c on ubuntu
Feb 12 13:14:10 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] default_access_ip_network_name = None {{(pid=1196) log_opt_
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 12 13:14:10 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] injected_network_template      = /opt/stack/nova/nova/virt/
interfaces.template {{(pid=1196) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 12 13:14:10 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] network_allocate_retries       = 0 {{(pid=1196) log_opt_val
ues /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 12 13:14:11 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] neutron.service_type           = network {{(pid=1196) log_o
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:11 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] scheduler.query_placement_for_routed_network_aggregates = F
alse {{(pid=1196) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:12 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] default_access_ip_network_name = None {{(pid=1195) log_opt_
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 12 13:14:12 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] injected_network_template      = /opt/stack/nova/nova/virt/
interfaces.template {{(pid=1195) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 12 13:14:12 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] network_allocate_retries       = 0 {{(pid=1195) log_opt_val
ues /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
```

Debug Events are displayed

Key Outcomes for Section 5.2:

- Successfully collected administrative evidence for users, roles, projects, networks, subnets, routers, security groups, and instance inventory.
- Verified that native OpenStack service logs and CLI output provide auditable evidence for identity, segmentation, and compute state validation.
- Demonstrated that Nova and Neutron activity can be correlated with previously implemented security controls.
- Confirmed that native services provide sufficient visibility to support security review and governance monitoring.

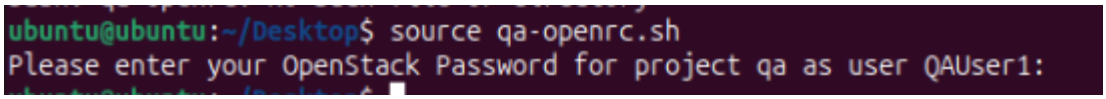
Section 5.3: Validation of Policy, Segmentation, and Governance Through Denied Events

Purpose

This section validates that security controls implemented in earlier reports—such as RBAC, tenant isolation, and quota enforcement—generate observable audit evidence when violated. The ability to detect denied or failed actions is critical for cloud security monitoring and governance [3].

This section validates that enforcement failures are both blocked and detectable. By intentionally attempting actions outside the QA user's authorized scope, the exercise confirms that governance controls generate reviewable evidence of policy violations, which mitigates the previously identified risk of unauthorized activity occurring without administrative visibility [3], [4].

Step 1: Source credentials for QA user



RC file sourced for QAUser1

Step 2: Attempt to create a network as QA user

```
ubuntu@ubuntu:~/Desktop$ openstack network create qa-test-network
+---------------------------+--------------------------------------+
| Field                     | Value                                |
+---------------------------+--------------------------------------+
| admin_state_up            | UP                                   |
| availability_zone_hints   |                                      |
| availability_zones        |                                      |
| created_at                | 2026-03-29T00:42:05Z                 |
| description               |                                      |
| dns_domain                | None                                 |
| id                        | b6581309-9b33-43cd-8e60-83f902be8bfc |
| ipv4_address_scope        | None                                 |
| ipv6_address_scope        | None                                 |
| is_default                | None                                 |
| is_vlan_qinq              | None                                 |
| is_vlan_transparent       | False                                |
| mtu                       | 1442                                 |
| name                      | qa-test-network                      |
| port_security_enabled     | True                                 |
| project_id                | 7e6bf84624f54b56b4da0f27b6c16aed     |
| provider:network_type     | None                                 |
| provider:physical_network | None                                 |
| provider:segmentation_id  | None                                 |
| qinq                      | False                                |
| qos_policy_id             | None                                 |
| revision_number           | 1                                    |
| router:external           | Internal                             |
| segments                  | None                                 |
| shared                    | False                                |
| status                    | ACTIVE                               |
| subnets                   |                                      |
| tags                      |                                      |
| updated_at                | 2026-03-29T00:42:05Z                 |
+---------------------------+--------------------------------------+
```

Step 3: Attempt to list all projects as QA user

```
ubuntu@ubuntu:~/Desktop$ openstack project list
+----------------------------------+------+
| ID                               | Name |
+----------------------------------+------+
| 7e6bf84624f54b56b4da0f27b6c16aed | qa   |
+----------------------------------+------+
```

Only the QA Project is viewable

Step 4: Attempt to view all instances

```
ubuntu@ubuntu:~/Desktop$ openstack server list --all-projects
ForbiddenException: 403: Client Error for url: http://127.0.0.1/compute/v2.1/servers/detail?deleted=False&all_tenants=True, Policy doesn't allow os_co
mpute_api:servers:detail:get_all_tenants to be performed.
```

The attempt to view all projects was denied

Step 5: Switch back to admin credentials

```
ubuntu@ubuntu:~$ source Desktop/admin-openrc.sh
Please enter your OpenStack Password for project admin as user admin:
ubuntu@ubuntu:~$
```

RC file sourced for admin

Step 6: Review Nova logs for denied or failed actions

```
ubuntu@ubuntu:~$ journalctl -u devstack@n-api.service | grep -i "denied|forbidden|unauthorized|policy|error|fail"
```

No matches were found

Step 7: Review Keystone-related authorization failures with the command: journalctl -u devstack@keystone.service | grep -i "fail"

```
Feb 13 17:10:13 ubuntu devstack@keystone.service[1185]: WARNING keystone.server.flask.application [None r
eq-d421110f-d6c7-494f-8927-80cfdf9bba12 None None] Authorization failed. The request you have made requir
es authentication. from 127.0.0.1: keystone.exception.Unauthorized: The request you have made requires au
thentication.
Feb 13 17:10:18 ubuntu devstack@keystone.service[1186]: WARNING keystone.server.flask.application [None r
eq-a489debc-8825-4384-80bc-58af316c704d None None] Authorization failed. The request you have made requir
es authentication. from 127.0.0.1: keystone.exception.Unauthorized: The request you have made requires au
thentication.
Feb 13 17:21:11 ubuntu devstack@keystone.service[1185]: WARNING keystone.server.flask.application [None r
eq-21b03cdf-ba86-4880-9d30-04b0a3d28cc4 None QAUser1] Authorization failed. The request you have made req
uires authentication. from 127.0.0.1: keystone.exception.Unauthorized: The request you have made requires
 authentication.
Feb 14 10:58:11 ubuntu devstack@keystone.service[1190]: DEBUG uwsgi [-] security_compliance.lockout_failu
re_attempts = 2 {{(pid=1190) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config
/cfg.py:2824}}
Feb 14 10:58:11 ubuntu devstack@keystone.service[1192]: DEBUG uwsgi [-] security_compliance.lockout_failu
re_attempts = 2 {{(pid=1192) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config
/cfg.py:2824}}
^C
ubuntu@ubuntu:~$
```

Keystone logs showing unauthorized access attempts

Step 8: Review Neutron logs for unauthorized network attempts

```
ubuntu@ubuntu:~$ journalctl -u devstack@neutron-api.service | grep -i "error"
Feb 11 19:45:17 ubuntu devstack@neutron-api.service[1136]: DEBUG neutron.common.config [-] logging_exception_prefix
  = ERROR %(name)s %(instance)s {{(pid=1136) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_confi
g/cfg.py:2817}}
Feb 11 19:45:17 ubuntu devstack@neutron-api.service[1136]: DEBUG neutron.common.config [-] publish_errors
  = False {{(pid=1136) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 11 19:45:17 ubuntu devstack@neutron-api.service[1136]: DEBUG neutron.common.config [-] profiler.filter_error_trace
  = False {{(pid=1136) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:18 ubuntu devstack@neutron-api.service[1135]: DEBUG neutron.common.config [-] logging_exception_prefix
  = ERROR %(name)s %(instance)s {{(pid=1135) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_confi
g/cfg.py:2817}}
Feb 11 19:45:18 ubuntu devstack@neutron-api.service[1135]: DEBUG neutron.common.config [-] publish_errors
  = False {{(pid=1135) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 11 19:45:19 ubuntu devstack@neutron-api.service[1135]: DEBUG neutron.common.config [-] profiler.filter_error_trace
  = False {{(pid=1135) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:42 ubuntu devstack@neutron-api.service[1136]: DEBUG oslo_db.sqlalchemy.engines [None req-62f0288b-c5f4-4af7
-8e40-a25f442a8d15 None None] MySQL server mode set to STRICT_TRANS_TABLES,STRICT_ALL_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DAT
E,ERROR_FOR_DIVISION_BY_ZERO,TRADITIONAL,NO_ENGINE_SUBSTITUTION {{(pid=1136) check_effective_sql_mode /opt/stack/data/v
```

Neutron service logs showing network-related errors

Step 9: Verify network isolation by listing networks per project

```
ubuntu@ubuntu:~/Desktop$ openstack network list --project qa
+--------------------------------------+-----------------+--------------------------------------+
| ID                                   | Name            | Subnets                              |
+--------------------------------------+-----------------+--------------------------------------+
| b6581309-9b33-43cd-8e60-83f902be8bfc | qa-test-network |                                      |
| d3e3e853-ec24-43aa-969b-1d7dfb5e9f63 | test-net        | 76cc7d70-112c-488d-acb8-5d1f99c5ca14 |
+--------------------------------------+-----------------+--------------------------------------+
```

The dev network is not seen in the QA project

```
ubuntu@ubuntu:~/Desktop$ openstack network list --project dev
+--------------------------------------+---------+--------------------------------------+
| ID                                   | Name    | Subnets                              |
+--------------------------------------+---------+--------------------------------------+
| a729b745-9ead-4256-b0b5-1ce16ef6d2f7 | dev-net | 9a8c2511-facf-4d12-bdaf-cb5a33ef1bec |
+--------------------------------------+---------+--------------------------------------+
```

The QA network is not seen in the dev project

Key Outcomes for Section 5.3:

- Successfully generated security-relevant denied-event activity by attempting actions outside the QA user context.
- Verified that RBAC and project-scoped restrictions limit access to administrative and cross-project resources.
- Demonstrated that failed and denied actions can be reviewed through Nova, Keystone, and Neutron log evidence.
- Confirmed that observable denied events improve accountability and mitigate the risk of silent policy violations.

Section 5.4: Administrative Review Workflow for Auditability and Anomaly Identification

Purpose

Administrators must be able to systematically review logs and system state to identify abnormal or policy-violating behavior. In a DevStack environment, this is achieved through centralized journald logs and OpenStack CLI inspection. Repeated failed actions, denied access attempts, and inconsistent resource usage patterns can indicate misuse or misconfiguration [4].

This section establishes a repeatable workflow for reviewing warning, error, authentication, networking, and compute events against the current cloud state. The results are intended to confirm that anomaly detection in the hardened environment can be performed using native tooling, strengthening auditability and reducing the likelihood that misconfiguration or abuse persists without review [1], [4].

Step 1: Review recent error logs across all DevStack services

```
ubuntu@ubuntu:~/Desktop$ journalctl -xe | grep -i "error"
Mar 28 20:53:48 ubuntu cinder-volume[4713]: ERROR cinder.service [-] Manager for service cinder-volume ubuntu@lvmdriver-1 is reporting problems, not s
ending heartbeat. Service will appear "down".
Mar 28 20:55:48 ubuntu cinder-volume[4713]: ERROR cinder.service [-] Manager for service cinder-volume ubuntu@lvmdriver-1 is reporting problems, not s
ending heartbeat. Service will appear "down".
Mar 28 20:57:48 ubuntu cinder-volume[4713]: ERROR cinder.service [-] Manager for service cinder-volume ubuntu@lvmdriver-1 is reporting problems, not s
ending heartbeat. Service will appear "down".
```

A common error only in the netlab environment. This is fixed each reboot.

Step 2: Review warning-level events with the command: journalctl -p warning --no-pager | tail -n 50

```
Mar 28 20:07:47 ubuntu systemd[7637]: app-gnome-gnome\x2dkeyring\x2dpkcs11-7864.scope: Failed with result 'resources'.
Mar 28 20:07:47 ubuntu systemd[7637]: Failed to start app-gnome-gnome\x2dkeyring\x2dpkcs11-7864.scope - Application launched by
Mar 28 20:07:47 ubuntu gnome-session-binary[7840]: GnomeDesktop-WARNING: Could not create transient scope for PID 7872: GDBus.E
us.Error.UnixProcessIdUnknown: Failed to set unit properties: No such process
Mar 28 20:07:49 ubuntu systemd[7637]: Dependency failed for org.gnome.SettingsDaemon.XSettings.service - GNOME XSettings servic
Mar 28 20:07:49 ubuntu gnome-shell[7884]: Gio.UnixInputStream has been moved to a separate platform-specific library. Please up
ioUnix.InputStream instead.
                                          0 inhibit() ["resource:///org/gnome/shell/misc/loginManager.js":209:8]
                                          1 InterpretGeneratorResume() ["self-hosted":1461:33]
                                          2 AsyncFunctionNext() ["self-hosted":852:26]
                                          3 anonymous() ["resource:///org/gnome/shell/ui/init.js":21:19]
Mar 28 20:07:49 ubuntu systemd[7637]: app-gnome-spice\x2dvdagent-8014.scope: No PIDs left to attach to the scope's control grou
Mar 28 20:07:49 ubuntu systemd[7637]: app-gnome-spice\x2dvdagent-8014.scope: Failed with result 'resources'.
Mar 28 20:07:49 ubuntu systemd[7637]: Failed to start app-gnome-spice\x2dvdagent-8014.scope - Application launched by gnome-ses
Mar 28 20:07:49 ubuntu at-spi2-registr[7956]: Failed to register client: GDBus.Error:org.gnome.SessionManager.AlreadyRegistered
ient
Mar 28 20:07:49 ubuntu at-spi2-registr[7956]: Unable to register client with session manager
Mar 28 20:07:49 ubuntu systemd[7637]: app-gnome-user\x2ddirs\x2dupdate\x2dgtk-8053.scope: No PIDs left to attach to the scope's
g.
Mar 28 20:07:49 ubuntu systemd[7637]: app-gnome-user\x2ddirs\x2dupdate\x2dgtk-8053.scope: Failed with result 'resources'.
Mar 28 20:07:49 ubuntu systemd[7637]: Failed to start app-gnome-user\x2ddirs\x2dupdate\x2dgtk-8053.scope - Application launched
y.
Mar 28 20:07:52 ubuntu gnome-shell[7157]: Connection to xwayland lost
Mar 28 20:07:52 ubuntu gnome-shell[7157]: Xwayland terminated, exiting since it was mandatory
Mar 28 20:07:52 ubuntu /usr/libexec/gdm-wayland-session[7108]: dbus-daemon[7108]: [session uid=120 pid=7108] Activating service
systemd1' requested by ':1.10' (uid=120 pid=7269 comm="/usr/libexec/gsd-sharing" label="unconfined")
Mar 28 20:07:52 ubuntu gsd-media-keys[8022]: Failed to grab accelerator for keybinding settings:hibernate
Mar 28 20:07:52 ubuntu gsd-media-keys[8022]: Failed to grab accelerator for keybinding settings:playback-repeat
Mar 28 20:07:52 ubuntu gdm3[6816]: Gdm: on_display_removed: assertion 'GDM_IS_REMOTE_DISPLAY (display)' failed
Mar 28 20:08:08 ubuntu gnome-shell[7884]: meta_window_set_stack_position_no_sync: assertion 'window->stack_position >= 0' faile
Mar 28 20:13:25 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:13:25 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:23:42 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:23:42 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:40:22 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:40:22 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:52:55 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
Mar 28 20:52:55 ubuntu update-notifier[8660]: gtk_widget_get_scale_factor: assertion 'GTK_IS_WIDGET (widget)' failed
ubuntu@ubuntu:~/Desktop$
```

Various Warnings are Displayed.

Step 3: Review authentication-related events with the command: journalctl | grep -i "auth"

```
core.dogpile=INFO', 'OFPHandler=INFO', 'OfctlService=INFO', 'os_ken.base.app_manager=INFO', 'os_ken.controller.controller=INFO'] {{(
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] oslo_middleware.http_basic_auth_user_file
(pid=1124) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] nova.auth_section              = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] nova.auth_type                = password
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] placement.auth_section        = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] placement.auth_type           = password
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] ironic.auth_section           = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1124]: DEBUG neutron.common.config [-] ironic.auth_type              = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] auth_strategy                 = keystone
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] default_log_levels            = ['amqp=WA
', 'boto=WARN', 'qpid=WARN', 'sqlalchemy=WARN', 'suds=INFO', 'oslo.messaging=INFO', 'oslo_messaging=INFO', 'iso8601=WARN', 'requests
connectionpool=WARN', 'urllib3.connectionpool=WARN', 'websocket=WARN', 'requests.packages.urllib3.util.retry=WARN', 'urllib3.util.re
nemiddleware=WARN', 'routes.middleware=WARN', 'stevedore=WARN', 'taskflow=WARN', 'keystoneauth=WARN', 'oslo.cache=INFO', 'oslo_polic
core.dogpile=INFO', 'OFPHandler=INFO', 'OfctlService=INFO', 'os_ken.base.app_manager=INFO', 'os_ken.controller.controller=INFO'] {{(
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] oslo_middleware.http_basic_auth_user_file
(pid=1123) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] nova.auth_section              = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] nova.auth_type                = password
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] placement.auth_section        = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] placement.auth_type           = password
pt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] ironic.auth_section           = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Jan 31 19:36:09 ubuntu devstack@neutron-api.service[1123]: DEBUG neutron.common.config [-] ironic.auth_type              = None {{(p
values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
```

The logs continue to display as many logs match the phrase "Auth"

Step 4: Review network-related anomalies with the command: journalctl -u devstack@n-api.service | grep -i "fail"

```
Feb 11 19:45:27 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] api.vendordata_dynamic_failure_fatal = False {{(pid=1132) l
og_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:29 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] filter_scheduler.build_failure_weight_multiplier = 1000000.
0 {{(pid=1132) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:29 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.enable_cancel_on_failover = False {{(
pid=1132) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:29 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.kombu_failover_strategy = round-robin
 {{(pid=1132) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:33 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] api.vendordata_dynamic_failure_fatal = False {{(pid=1133) l
og_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:35 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] filter_scheduler.build_failure_weight_multiplier = 1000000.
0 {{(pid=1133) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:35 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.enable_cancel_on_failover = False {{(
pid=1133) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:35 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.kombu_failover_strategy = round-robin
 {{(pid=1133) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:57 ubuntu devstack@n-api.service[1132]: ERROR dbcounter [-] Failed to account for access to database 'nova_api': (pymysql.err.InternalErr
or) Packet sequence number wrong - got 1 expected 2
Feb 12 13:14:10 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] api.vendordata_dynamic_failure_fatal = False {{(pid=1196) l
og_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:11 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] filter_scheduler.build_failure_weight_multiplier = 1000000.
0 {{(pid=1196) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:12 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.enable_cancel_on_failover = False {{(
pid=1196) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:12 ubuntu devstack@n-api.service[1196]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.kombu_failover_strategy = round-robin
 {{(pid=1196) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:13 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] api.vendordata_dynamic_failure_fatal = False {{(pid=1195) l
og_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:14 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] filter_scheduler.build_failure_weight_multiplier = 1000000.
0 {{(pid=1195) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:14 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.enable_cancel_on_failover = False {{(
pid=1195) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 12 13:14:14 ubuntu devstack@n-api.service[1195]: DEBUG nova.api.openstack.wsgi_app [-] oslo_messaging_rabbit.kombu_failover_strategy = round-robin
 {{(pid=1195) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
```

You will notice that configurations regarding failure also appear here.

Step 5: Review compute-related failures with the command: journalctl -u devstack@n-api.service | grep -i "error"

```
(pid=1117) _check_effective_sql_mode /opt/stack/data/venv/lib/python3.12/site-packages/oslo_db/sqlalchemy/engines.py:397}}
Jan 31 19:50:47 ubuntu devstack@n-api.service[1117]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: Server unexpectedly closed connection
Jan 31 19:50:47 ubuntu devstack@n-api.service[1114]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: Server unexpectedly closed connection
Jan 31 20:11:15 ubuntu devstack@n-api.service[1114]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: [Errno 104] Connection reset by peer
Jan 31 20:11:16 ubuntu devstack@n-api.service[1117]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: [Errno 104] Connection reset by peer
Jan 31 20:11:16 ubuntu devstack@n-api.service[1117]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: [Errno 104] Connection reset by peer
Jan 31 20:19:32 ubuntu devstack@n-api.service[1114]: INFO oslo.messaging._drivers.impl_rabbit [-] A recoverable connection/channel error occurred, try
ing to reconnect: [Errno 104] Connection reset by peer
Feb 11 19:45:27 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] logging_exception_prefix       = ERROR %(name)s %(instance)
s {{(pid=1132) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 11 19:45:27 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] publish_errors                 = False {{(pid=1132) log_opt
_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 11 19:45:29 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] oslo_versionedobjects.fatal_exception_format_errors = False
 {{(pid=1132) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:29 ubuntu devstack@n-api.service[1132]: DEBUG nova.api.openstack.wsgi_app [-] profiler.filter_error_trace    = False {{(pid=1132) log_opt
_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2824}}
Feb 11 19:45:30 ubuntu devstack@n-api.service[1132]: DEBUG oslo_db.sqlalchemy.engines [None req-1ed0c449-3707-4fe5-b5cb-28d04ea92c2c None None] MySQL
server mode set to STRICT_TRANS_TABLES,STRICT_ALL_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,TRADITIONAL,NO_ENGINE_SUBSTITUTION {{
(pid=1132) _check_effective_sql_mode /opt/stack/data/venv/lib/python3.12/site-packages/oslo_db/sqlalchemy/engines.py:397}}
Feb 11 19:45:32 ubuntu devstack@n-api.service[1132]: DEBUG oslo_db.sqlalchemy.engines [None req-1ed0c449-3707-4fe5-b5cb-28d04ea92c2c None None] MySQL
server mode set to STRICT_TRANS_TABLES,STRICT_ALL_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,TRADITIONAL,NO_ENGINE_SUBSTITUTION {{
(pid=1132) _check_effective_sql_mode /opt/stack/data/venv/lib/python3.12/site-packages/oslo_db/sqlalchemy/engines.py:397}}
Feb 11 19:45:32 ubuntu devstack@n-api.service[1132]: DEBUG oslo_db.sqlalchemy.engines [None req-1ed0c449-3707-4fe5-b5cb-28d04ea92c2c None None] MySQL
server mode set to STRICT_TRANS_TABLES,STRICT_ALL_TABLES,NO_ZERO_IN_DATE,NO_ZERO_DATE,ERROR_FOR_DIVISION_BY_ZERO,TRADITIONAL,NO_ENGINE_SUBSTITUTION {{
(pid=1132) _check_effective_sql_mode /opt/stack/data/venv/lib/python3.12/site-packages/oslo_db/sqlalchemy/engines.py:397}}
Feb 11 19:45:32 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] logging_exception_prefix       = ERROR %(name)s %(instance)
s {{(pid=1133) log_opt_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
Feb 11 19:45:33 ubuntu devstack@n-api.service[1133]: DEBUG nova.api.openstack.wsgi_app [-] publish_errors                 = False {{(pid=1133) log_opt
_values /opt/stack/data/venv/lib/python3.12/site-packages/oslo_config/cfg.py:2817}}
```

Various Devstack errors are displayed.

Step 6: Verify current network state remains consistent

```
ubuntu@ubuntu:~/Desktop$ openstack network list
+--------------------------------------+----------------+----------------------------------------------------------------------------+
| ID                                   | Name           | Subnets                                                                    |
+--------------------------------------+----------------+----------------------------------------------------------------------------+
| 24353f0c-c770-4e7a-abd7-196dcb3bc476 | public         | 0fd39408-8388-411b-9d1a-f564cfd67967, 95515952-8b36-4743-924d-21b00e17e63c |
| 398b5405-9388-4b39-9937-8e855b33c735 | shared         | fd117f77-351a-4a10-a454-beb9919beca3                                       |
| a729b745-9ead-4256-b0b5-1ce16ef6d2f7 | dev-net        | 9a8c2511-facf-4d12-bdaf-cb5a33ef1bec                                       |
| b6581309-9b33-43cd-8e60-83f902be8bfc | qa-test-network |                                                                            |
| d3e3e853-ec24-43aa-969b-1d7dfb5e9f63 | test-net       | 76cc7d70-112c-488d-acb8-5d1f99c5ca14                                       |
| e7eefaea-083b-4004-bf10-1aa9c76172d5 | private        | f77829e9-5891-42f9-a3bf-689666dc902c, fa40244b-2d8b-4e48-bbce-a23a60a240c1 |
+--------------------------------------+----------------+----------------------------------------------------------------------------+
```

All networks appear as expected.

```
ubuntu@ubuntu:~/Desktop$ openstack subnet list
+--------------------------------------+---------------------+--------------------------------------+---------------------+
| ID                                   | Name                | Network                              | Subnet              |
+--------------------------------------+---------------------+--------------------------------------+---------------------+
| 0fd39408-8388-411b-9d1a-f564cfd67967 | ipv6-public-subnet  | 24353f0c-c770-4e7a-abd7-196dcb3bc476 | 2001:db8::/64       |
| 76cc7d70-112c-488d-acb8-5d1f99c5ca14 | test-subnet         | d3e3e853-ec24-43aa-969b-1d7dfb5e9f63 | 10.0.10.0/24        |
| 95515952-8b36-4743-924d-21b00e17e63c | public-subnet       | 24353f0c-c770-4e7a-abd7-196dcb3bc476 | 172.24.4.0/24       |
| 9a8c2511-facf-4d12-bdaf-cb5a33ef1bec | dev-subnet          | a729b745-9ead-4256-b0b5-1ce16ef6d2f7 | 10.0.20.0/24        |
| f77829e9-5891-42f9-a3bf-689666dc902c | private-subnet      | e7eefaea-083b-4004-bf10-1aa9c76172d5 | 10.0.0.0/26         |
| fa40244b-2d8b-4e48-bbce-a23a60a240c1 | ipv6-private-subnet | e7eefaea-083b-4004-bf10-1aa9c76172d5 | fd59:3f9e:7361::/64 |
| fd117f77-351a-4a10-a454-beb9919beca3 | shared-subnet       | 398b5405-9388-4b39-9937-8e855b33c735 | 192.168.233.0/24    |
+--------------------------------------+---------------------+--------------------------------------+---------------------+
```

All subnets appear as expected.

```
ubuntu@ubuntu:~/Desktop$ openstack router list
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
| ID                                   | Name        | Status | State | Project                          | HA   |
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
| 077bdf3b-2cd5-45d3-9156-8a5aaa4704d5 | dev-router  | ACTIVE | UP    | 4f97f0eebec74d8b83b9296bc70dd0bb | True |
| 1ab77af1-4929-4525-8c72-1da8453f7bbc | router1     | ACTIVE | UP    | ecfddf1248db4233aad57146f0b0b792 | True |
| d1019b36-f21b-490b-84ae-3c6ccb51b52d | test-router | ACTIVE | UP    | 7e6bf84624f54b56b4da0f27b6c16aed | True |
+--------------------------------------+-------------+--------+-------+----------------------------------+------+
```

All routers are listed as expected.

Step 7: Verify current security group configuration

```
ubuntu@ubuntu:~/Desktop$ openstack security group list
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
| ID                                   | Name        | Description                 | Project                          | Tags | Shared |
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
| 1c53e01b-9b6b-431b-8b12-964836832e18 | test-secure | TEST inbound: SSH, ICMP only | 7e6bf84624f54b56b4da0f27b6c16aed | []   | False  |
| 278ac93e-b084-418e-8636-426c452fefd1 | dev-secure  | DEV inbound: SSH, ICMP, HTTP | 4f97f0eebec74d8b83b9296bc70dd0bb | []   | False  |
| 3ca27139-4e0d-41fa-9c43-9f8ee7f0312d | default     | Default security group      | ecfddf1248db4233aad57146f0b0b792 | []   | False  |
| 8fb9e636-2eae-4b42-a99c-84ea50dbb2c0 | default     | Default security group      | fd2553c6119d4c41aae1a732d85d9107 | []   | False  |
| a6f23c36-121e-4c3e-9a7f-fc1870b81aef | default     | Default security group      | 4f97f0eebec74d8b83b9296bc70dd0bb | []   | False  |
| ac040561-2ce1-43fb-8f5f-6551f53fd301 | default     | Default security group      | 7e6bf84624f54b56b4da0f27b6c16aed | []   | False  |
| ba8bc8c6-7f60-4d7f-9a0c-1b8870315774 | default     | Default security group      | bd755daf91444d328d3f25316317a883 | []   | False  |
+--------------------------------------+-------------+-----------------------------+----------------------------------+------+--------+
```

All Security Groups are reported as expected.

Step 8: Verify instance state across projects

```
ubuntu@ubuntu:~/Desktop$ openstack server list --all-projects
+----------------------+----------------------+--------+----------------------+----------------------+---------+----------------------+
| ID                   | Name                 | Status | Networks             | Image                | Flavor  | Project ID           |
+----------------------+----------------------+--------+----------------------+----------------------+---------+----------------------+
| 6dd37da6-1bc3-4364-  | DEV-APP              | ACTIVE | dev-net=10.0.20.86   | cirros-0.6.3-x86_64- | m1.tiny | 4f97f0eebec74d8b83b9296bc7 |
| 856f-510968779773    |                      |        |                      | disk                 |         | 0dd0bb               |
| d0580bec-717d-40ac-  | governance-test-     | SHUTOFF | dev-net=10.0.20.71   | cirros-0.6.3-x86_64- | m1.tiny | 4f97f0eebec74d8b83b9296bc7 |
| 877e-99509090437f    | instance             |        |                      | disk                 |         | 0dd0bb               |
| ec9d8b0f-b1b0-4166-  | TEST-DB              | SHUTOFF | test-net=10.0.10.195, | N/A (booted from     | m1.tiny | 7e6bf84624f54b56b4da0f27b6 |
| a0e7-60688f7bb11c    |                      |        | 172.24.4.107         | volume)              |         | c16aed               |
| 0925a67e-0b77-4536-  | TEST-APP             | SHUTOFF | test-net=10.0.10.184, | N/A (booted from     | m1.tiny | 7e6bf84624f54b56b4da0f27b6 |
| 8789-5f8207605940    |                      |        | 172.24.4.18          | volume)              |         | c16aed               |
| f45bc2de-808d-46ab-  | DEV-DB               | SHUTOFF | dev-net=10.0.20.126, | N/A (booted from     | m1.tiny | 4f97f0eebec74d8b83b9296bc7 |
| bce3-294d37c939bb    |                      |        | 172.24.4.78          | volume)              |         | 0dd0bb               |
+----------------------+----------------------+--------+----------------------+----------------------+---------+----------------------+
```

All servers are reported as expected.

Key Outcomes for Section 5.4:

- Successfully reviewed centralized log data for error, warning, authentication, network, and compute anomalies.
- Verified that current network, security group, and instance state remains aligned with the hardened administrative baseline.
- Demonstrated a repeatable administrative workflow for identifying policy violations, misuse indicators, and misconfiguration.
- Confirmed that structured log review strengthens auditability, governance, and ongoing security validation.

Engagement Summary

This engagement demonstrates that the hardened OpenStack environment now supports auditable validation in addition to preventive control enforcement. Through native service logs and administrative state inspection, administrators can review whether identity restrictions, network segmentation, and governance controls remain observable, enforceable, and suitable for ongoing security oversight.

- Established a baseline audit surface across critical DevStack-managed OpenStack services.
- Validated collection of identity, network, and compute evidence through native OpenStack interfaces.
- Demonstrated that denied events provide measurable evidence of RBAC, segmentation, and governance enforcement.
- Confirmed that a structured administrative review process improves accountability and anomaly detection.

References

[1] OpenStack Foundation, “Operations Guide,” OpenStack Documentation, 2024. [Online]. Available: https://docs.openstack.org/operations-guide/

[2] OpenStack Foundation, “Logging and Monitoring,” OpenStack Documentation, 2024. [Online]. Available: https://docs.openstack.org/operations-guide/ops-monitoring.html

[3] OpenStack Foundation, “Identity Service (Keystone) and Role-Based Access Control (RBAC),” OpenStack Documentation, 2024. [Online]. Available: https://docs.openstack.org/keystone/latest/admin/

[4] OpenStack Foundation, “OpenStack Security Guide,” OpenStack Documentation, 2024. [Online]. Available: https://docs.openstack.org/security-guide/